



## Department of Chemistry

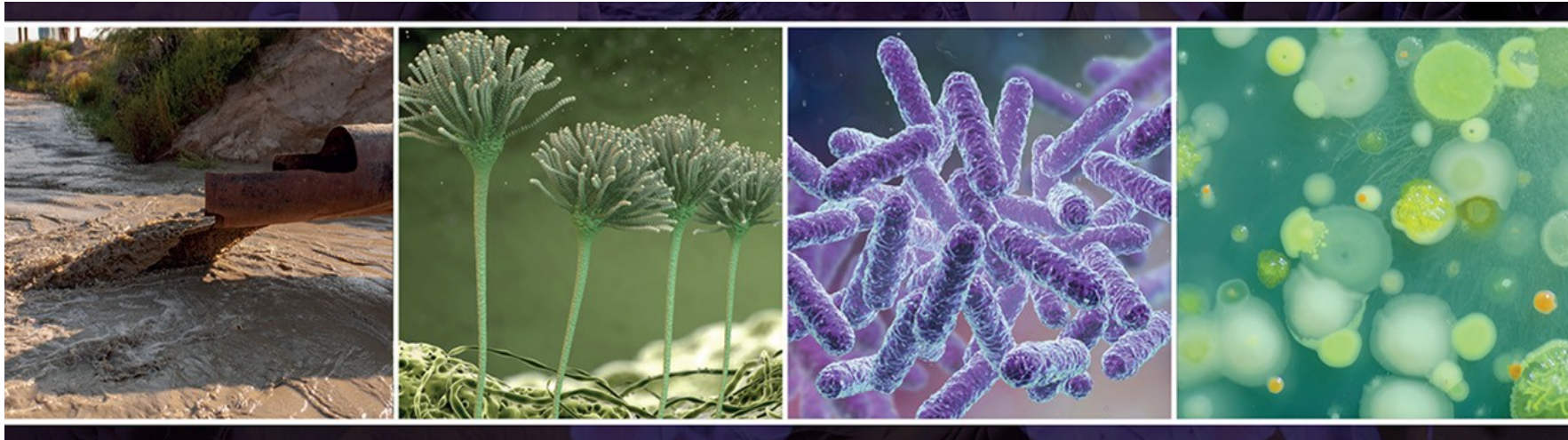
**Environmental Chemistry (2 credit)**

**CY1010**

July 2026

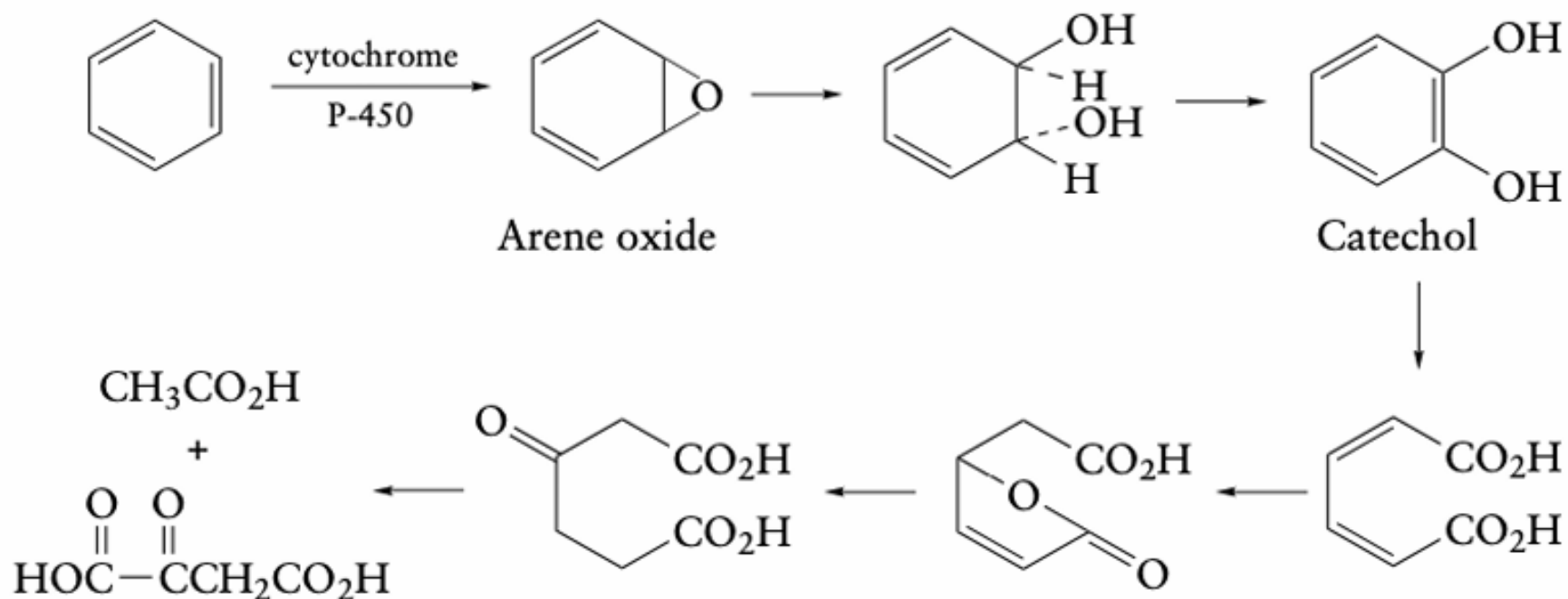
# Course Content

**Part 2 (Dr Abhijit): Aspects of transformations in atmosphere (microbial degradation of organics, environmental degradation of polymers, atmospheric lifetime, toxicity). Green chemistry and industrial Ecology.**



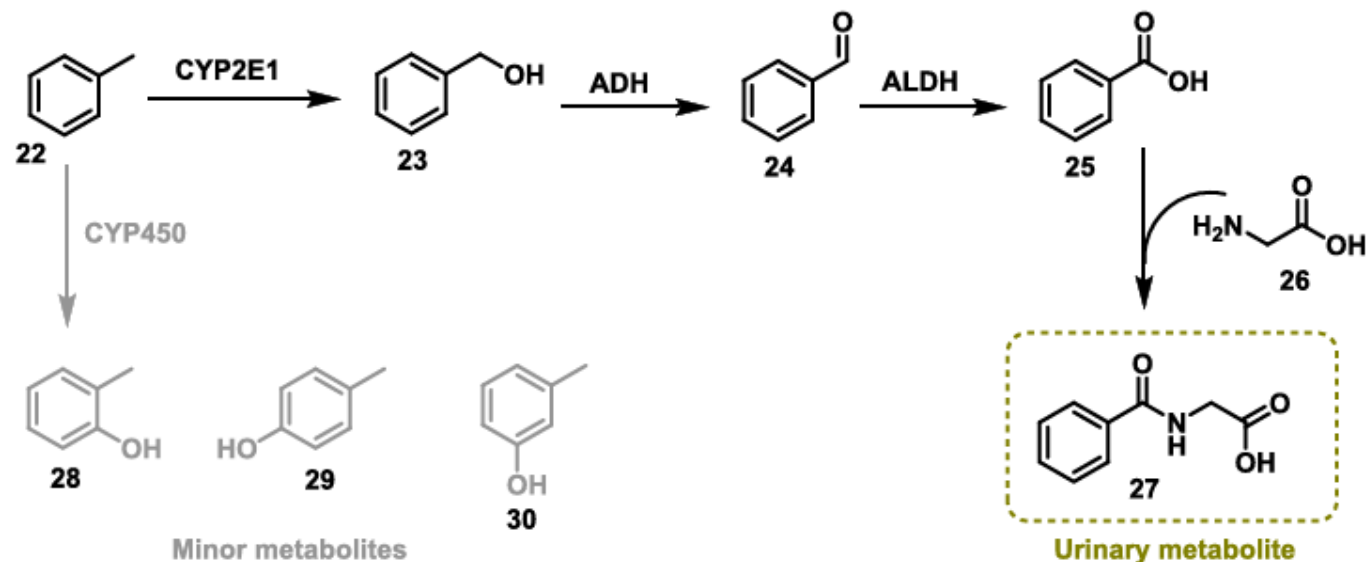
# Oxidation of the cyclic hydrocarbons and aromatic compounds

The microbial oxidation of the cyclic hydrocarbons and aromatic compounds present in detergents and petroleum is also catalysed by the enzyme cytochrome P-450 in microorganisms.



# Oxidation of the cyclic hydrocarbons and aromatic compounds

Scheme 3. Toluene Metabolism

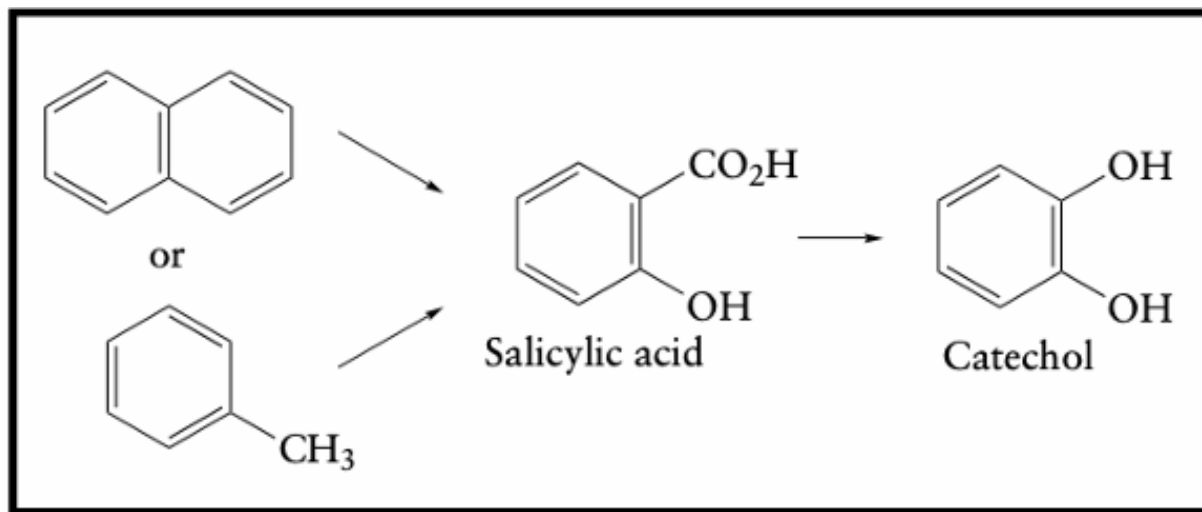


Hippuric acid is biosynthesized from benzoic acid and glycine in the liver, intestine, and kidneys, with benzoic acid first converted to benzoyl-CoA, an activated acylating agent that subsequently reacts with glycine to form hippuric acid.

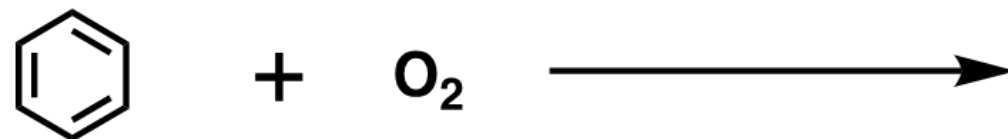
# Oxidation of the cyclic hydrocarbons and aromatic compounds

The microbial oxidation of the cyclic hydrocarbons and aromatic compounds present in detergents and petroleum is also catalysed by the enzyme cytochrome P-450 in microorganisms.

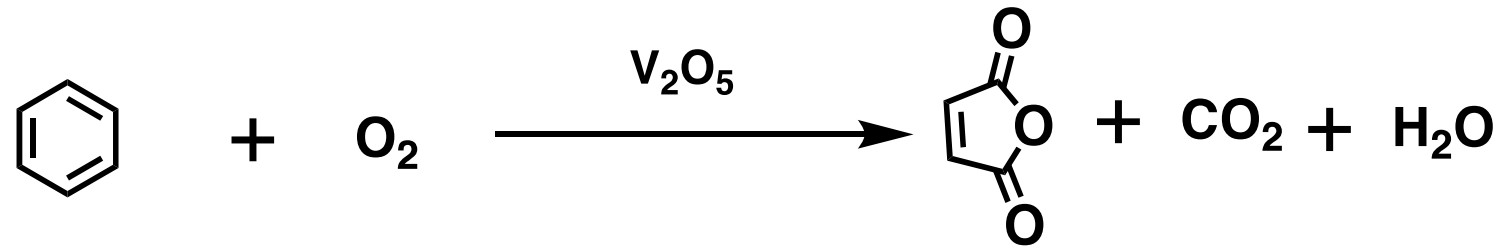
more highly condensed aromatics  
proceeds via salicylic acid



# Chemical Oxidation of Benzene



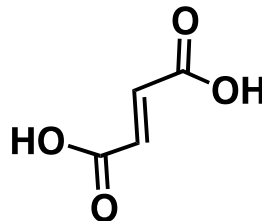
# Chemical Oxidation of Benzene



**Oxidation of benzene produce maleic anhydride**

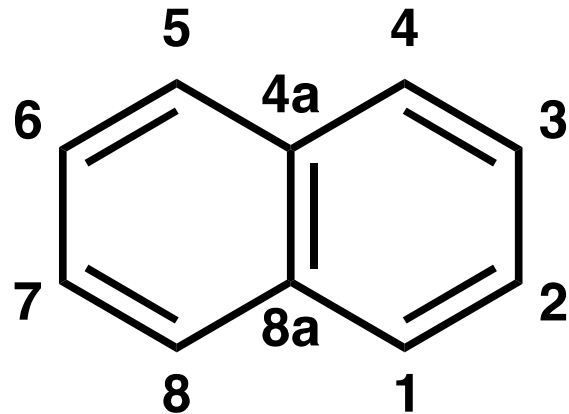
**Benzene and air mixture is oxidized to maleic anhydride over the catalyst at 2-5 bar, 400-450 °C,**

**Byproducts:**



**Fumaric acid**

# Oxidation Products of Naphthalene

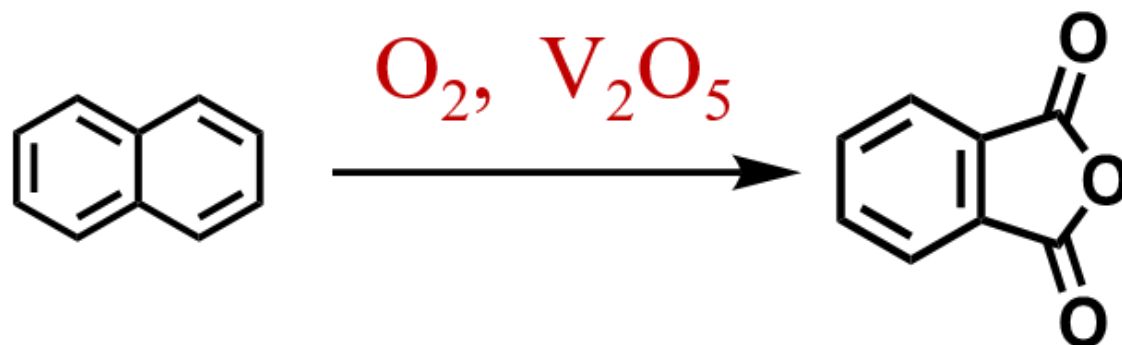


**Naphthalene:** Two benzene rings are fused in ortho positions and isolation from coal-tar

It is a white solid having a strong odour (m.p.  $80^{\circ}\text{C}$ , b.p.  $218^{\circ}\text{C}$ )

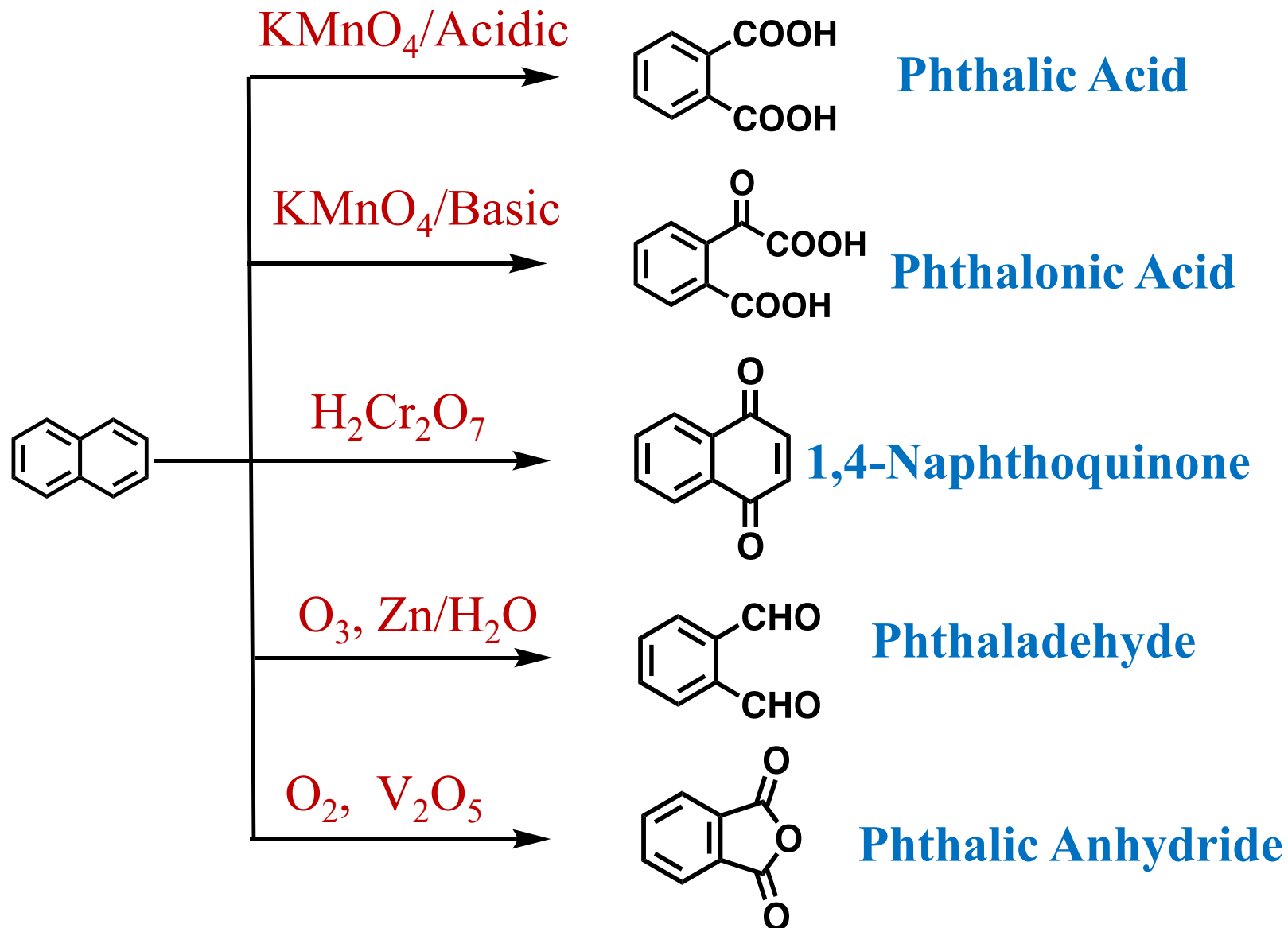


# Oxidation Products of Naphthalene

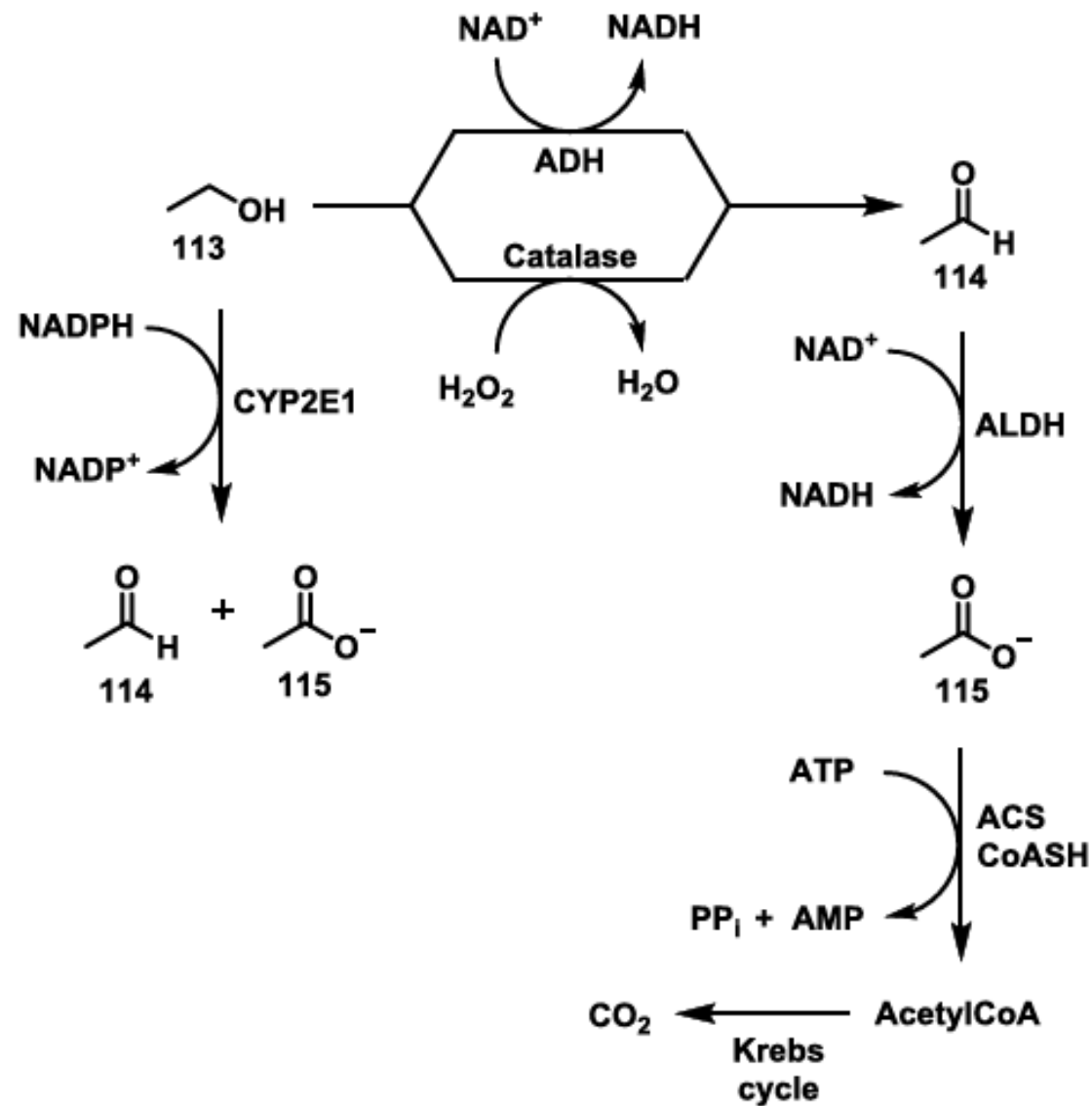


**Phthalic Anhydride**

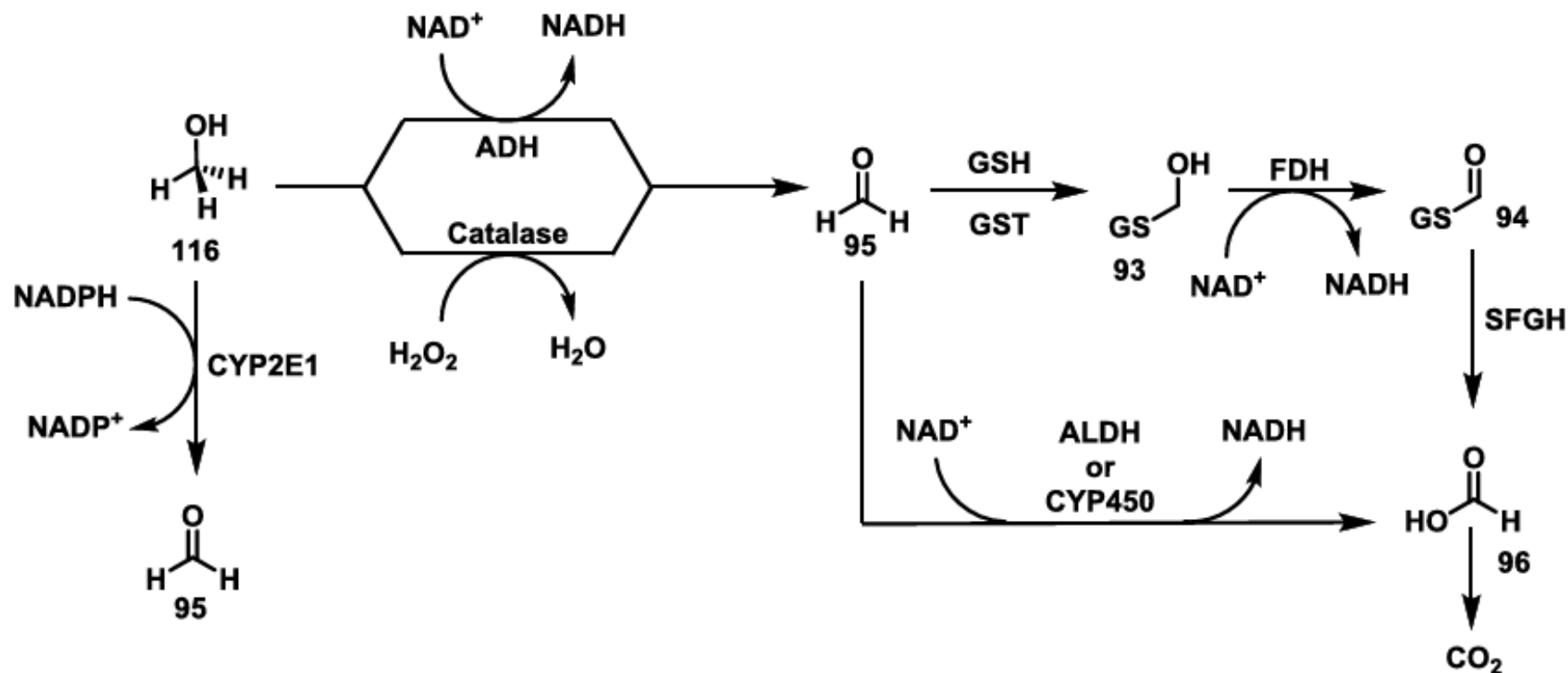
# Oxidation Products of Naphthalene



Scheme 15. Ethanol Metabolism (ACS = Acetyl-CoA Synthetase)



Scheme 16. Methanol Metabolism (SFGH = S-Formylglutathione Hydrolase)

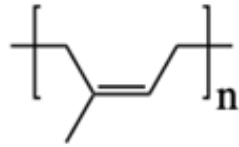


The first metabolite formed during methanol oxidation is formaldehyde (95), which is responsible for neuronal damage in the brain and other toxic effects.

## *Microbial degradation of organics*

## *Environmental degradation of polymers*

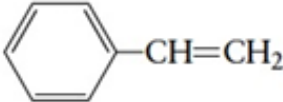
Natural rubber is not easily biodegradable



**cis-1,4- polyisoprene**

**Proteins:** polymers of amino acids

**Starch and Cellulose:** polymers of sugars

| Polymer       | Repeat unit                                                                 | Monomer(s)                                                                           |
|---------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Polyethylene  | $-\text{CH}_2\text{CH}_2-$                                                  | $\text{H}_2\text{C}=\text{CH}_2$                                                     |
| Polypropylene | $\begin{array}{c} -\text{CHCH}_2- \\   \\ \text{CH}_3 \end{array}$          | $\text{CH}_3\text{HC}=\text{CH}_2$                                                   |
| Polystyrene   | $\begin{array}{c} -\text{CHCH}_2- \\   \\ \text{C}_6\text{H}_5 \end{array}$ |  |

- Globally, only 18% of plastics waste are recycled, and 24% are incinerated
- 58% is placed in landfills (dumping yards)
- Relatively little decomposition takes place in landfills, so very little of this plastic material disappears with time (stable for more than 500 years)

# Life cycle of polymeric materials and main recycling technologies.

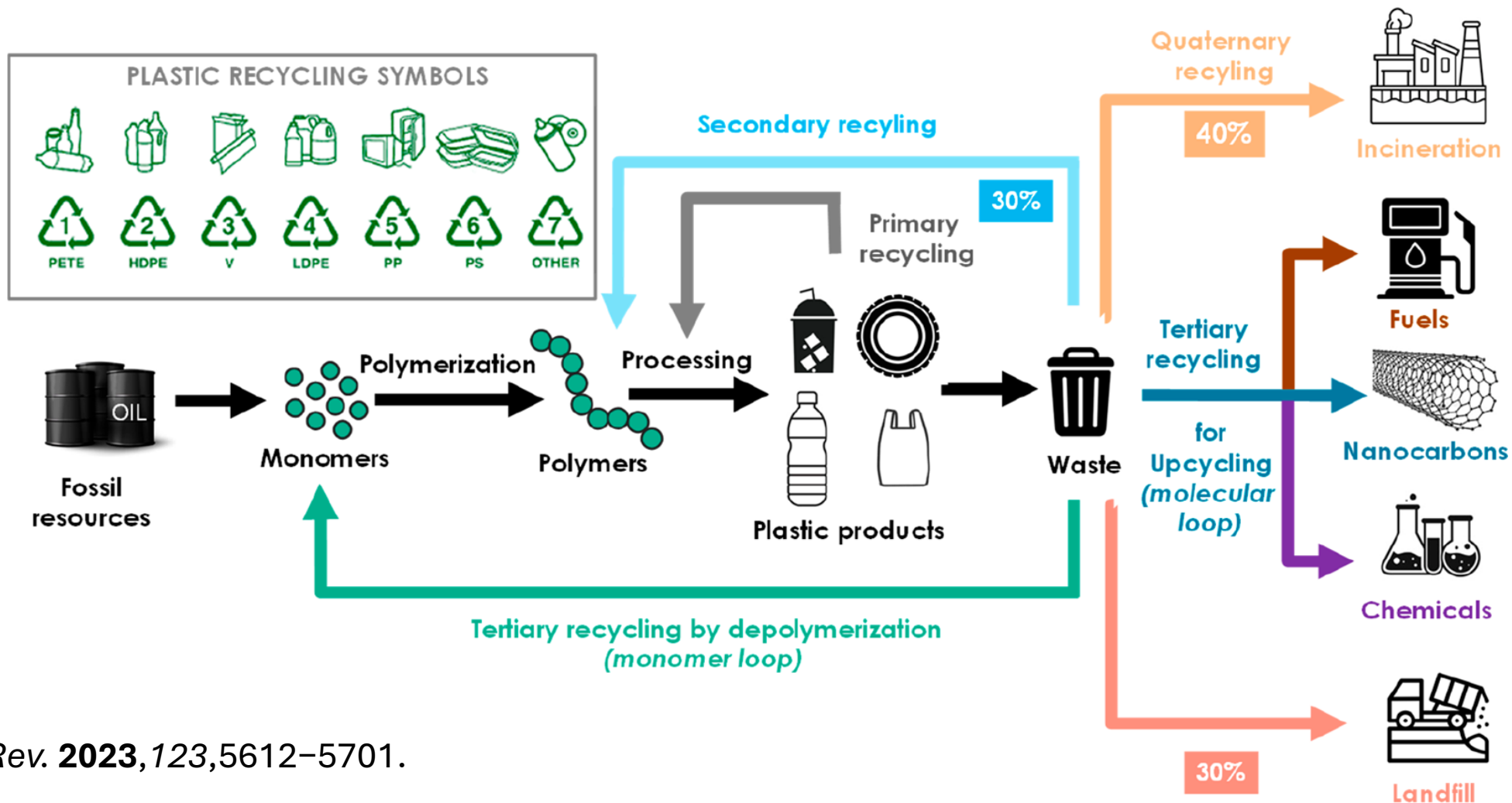
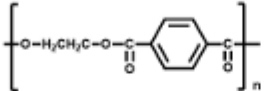
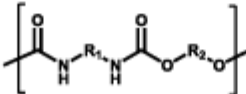
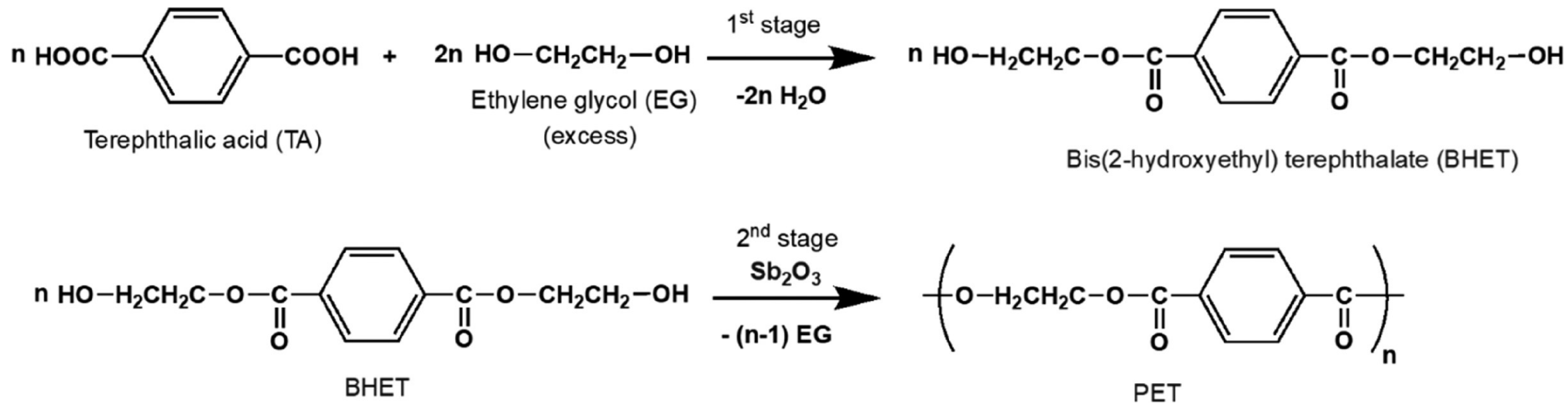


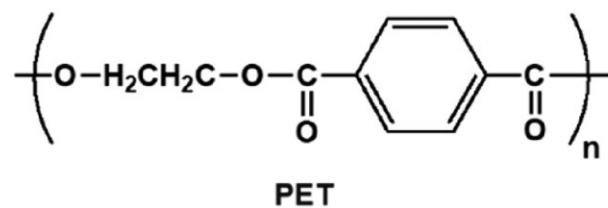
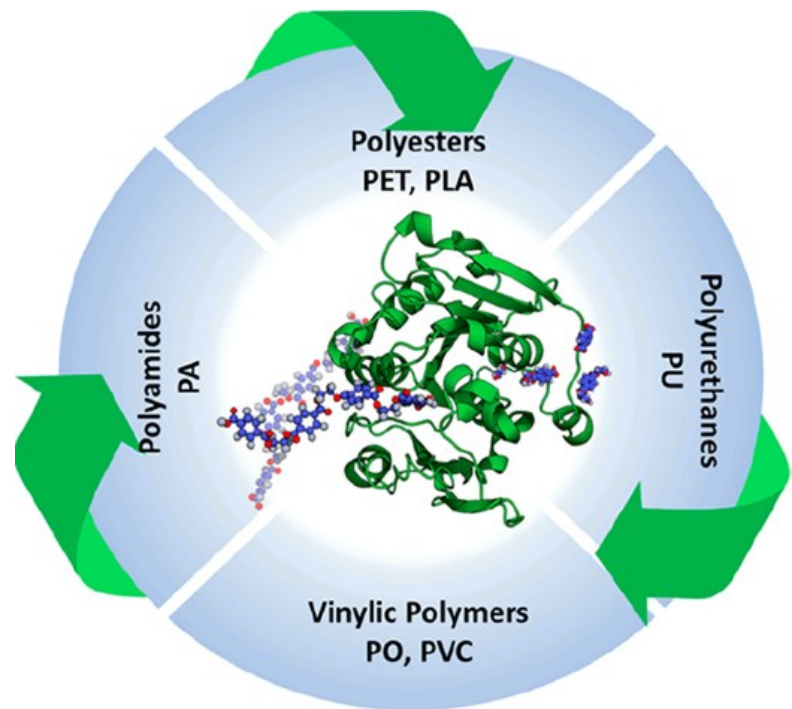
Table 1. Polymer Constituting the Main Plastics to Be Degraded

| Polymer                                             | Abbreviation and code | Chemical Structure                                                                   | Major uses; common commercial products                                                        | $T_g$ (°C) <sup>a</sup> | $T_m$ (°C) <sup>b</sup> | Density (g/cm <sup>3</sup> ) |
|-----------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------|-------------------------|------------------------------|
| Poly(ethylene terephthalate)                        | PET<br>1              |    | Packaging food and beverages; drink bottles, cups, food containers, textile fibers            | 80                      | > 250                   | 1.38–1.40                    |
| High-density polyethylene                           | HDPE<br>2             |   | Packaging; grocery bags; containers for shampoo, motor oil, detergent bleach                  | ~ -100                  | 135                     | 0.93–0.97                    |
| Poly(vinyl chloride)                                | PVC<br>3              |   | Building; pipeline tubes, cables, garden furniture, fencing and carpet backing                | 60–70                   | - <sup>c</sup>          | 1.10–1.45                    |
| Low-density polyethylene                            | LDPE<br>4             |   | Packaging; bags, wrapping films, trays, computer components                                   | ~ -100                  | 90–110                  | 0.91–0.94                    |
| Polypropylene                                       | PP<br>5               |   | Packaging; bottle caps, luggage, dishware, furniture, appliances, car parts                   | -15–18                  | 130–170                 | 0.90–0.92                    |
| Polystyrene                                         | PS<br>6               |  | Packaging and building; carryout containers, trays, cups, foam packaging                      | 60–110                  | -                       | 0.96–1.04                    |
| Polyurethanes<br>Polycarbonates,<br>Polyesters, ... | Others<br>7           |  | Diverse applications; Multilayer packaging, some food containers, CD's, DVD's, safety glasses | Various                 | Various                 |                              |

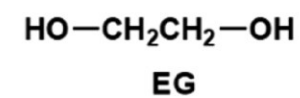
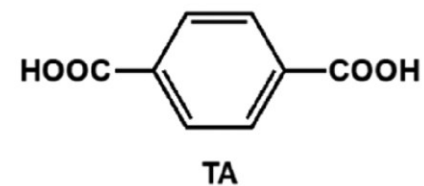
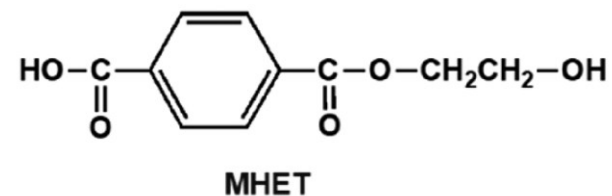
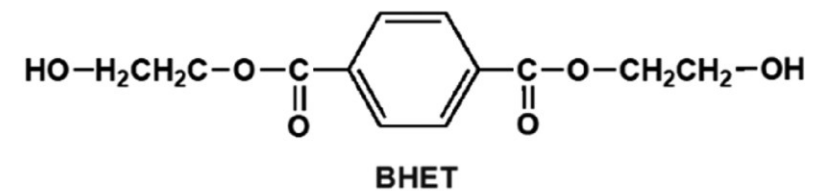
<sup>a</sup> $T_g$ : glass transition temperature <sup>b</sup> $T_m$ : melting temperature <sup>c</sup>The  $T_m$  value of commercial PVC can range from 110 °C to 250–260 °C, depending on the size of the crystallites. The melting and processing behavior of PVC have been discussed in ref 14.



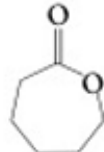




PET hydrolase



## Some typical polymers that are biodegradable under aerobic conditions

| Polymer                         | Repeat unit                                                                                                          | Monomer(s)                                                                                                                 |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Poly(vinyl acetate)             | $\begin{array}{c} \text{CH}_3 \\   \\ \text{C}=\text{O} \\   \\ \text{O} \\   \\ -\text{CH}_2\text{CH}- \end{array}$ | $\begin{array}{c} \text{CH}_3 \\   \\ \text{C}=\text{O} \\   \\ \text{O} \\   \\ \text{H}_2\text{C}=\text{CH} \end{array}$ |
| Poly(vinyl alcohol)             | $\begin{array}{c} \text{OH} \\   \\ -\text{CH}_2\text{CH}- \end{array}$                                              |                                                                                                                            |
| Poly(ethylene oxide)            | $-\text{CH}_2\text{CH}_2\text{O}-$                                                                                   | $\begin{array}{c} \text{O} \\ \diagup \quad \diagdown \\ \text{H}_2\text{C}-\text{CH}_2 \end{array}$                       |
| Polyglycolate                   | $\begin{array}{c} \text{O} \\    \\ -\text{OCH}_2\text{C}- \end{array}$                                              | $\text{HOCH}_2\text{CO}_2\text{H}$                                                                                         |
| Poly(lactate)                   | $\begin{array}{c} \text{O} \\    \\ -\text{OCHC}- \\   \\ \text{CH}_3 \end{array}$                                   | $\begin{array}{c} \text{HOCHCO}_2\text{H} \\   \\ \text{CH}_3 \end{array}$                                                 |
| Poly( $\beta$ -hydroxybutyrate) | $\begin{array}{c} \text{O} \\    \\ -\text{OCHCH}_2\text{C}- \\   \\ \text{CH}_3 \end{array}$                        | $\begin{array}{c} \text{HOCHCH}_2\text{CO}_2\text{H} \\   \\ \text{CH}_3 \end{array}$                                      |
| Poly( $\beta$ -hydroxyvalerate) | $\begin{array}{c} \text{O} \\    \\ -\text{OCHCH}_2\text{C}- \\   \\ \text{CH}_2\text{CH}_3 \end{array}$             | $\begin{array}{c} \text{HOCHCH}_2\text{CO}_2\text{H} \\   \\ \text{CH}_2\text{CH}_3 \end{array}$                           |
| Polycaprolactone                | $-\text{O}(\text{CH}_2)_5\text{C}(=\text{O})-$                                                                       |                                       |

Poly  $\beta$ -hydroxybutyrate (PHB) and the copolymer of  $\beta$ -hydroxybutyrate and  $\beta$ -hydroxyvalerate (PHBV) are unique in that they are synthesized by microorganisms

It has been possible to induce the microorganisms to produce the copolymer PHVB by adding  $\beta$ -hydroxyvalerate to their growth medium.

In 1999 it was reported that a genetically engineered plant (oilseed rape) produced PHVB directly. This process is not economically feasible at present.

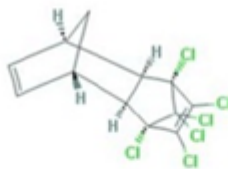
A polyethylene containing 5-20% starch granules is used in the manufacture of shopping bags that are claimed to be biodegradable.

Other additives such as transition metals may also be added to the polyethylene to enhance the photodegradation of the polyethylene via hydroperoxides

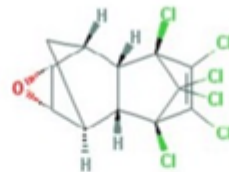
# **Organic Chemicals in the Environment**

# Organic Chemicals in the Environment

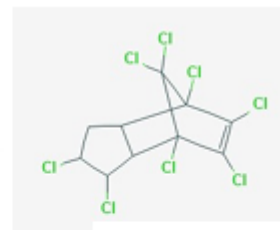
- pharmaceutical agents
  - Fibers
  - building materials
  - agricultural chemicals
  - solvents
  - cleaning agents
- inexpensive to manufacture
- used in large quantities



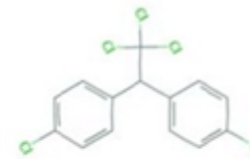
Aldrin



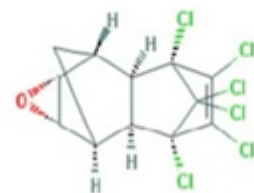
Dieldrin



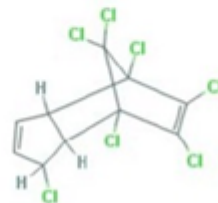
Chlordane



DDT



Endrin



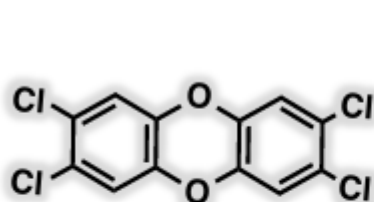
Heptachlor



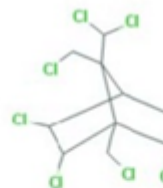
HCB



Mirex



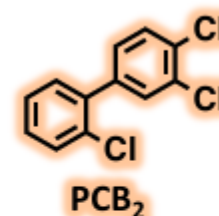
Dioxins



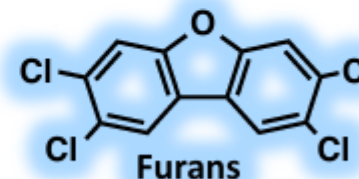
Toxaphene



PCB<sub>1</sub>



PCB<sub>2</sub>



Furans

Polychlorinated compounds

2,3,7,8-tetrachlorodibenzo[b,d]furan

# Organic Chemicals in the Environment

- **Pharmaceutical agents** – medicines, antibiotics, hormones, etc.
- **Fibers** – synthetic fibers such as polyester, nylon, and acrylic.
- **Building materials** – paints, adhesives, plastics, sealants, and insulation materials.
- **Agricultural chemicals** – pesticides, herbicides, fungicides, and insecticides.
- 
- **Solvents** – benzene, toluene, acetone, and chlorinated solvents.
- **Cleaning agents** – detergents, disinfectants, degreasers, and surfactants.

# The 12 Key Persistent Organic Pollutants (POPs)

Stockholm Convention In 2001

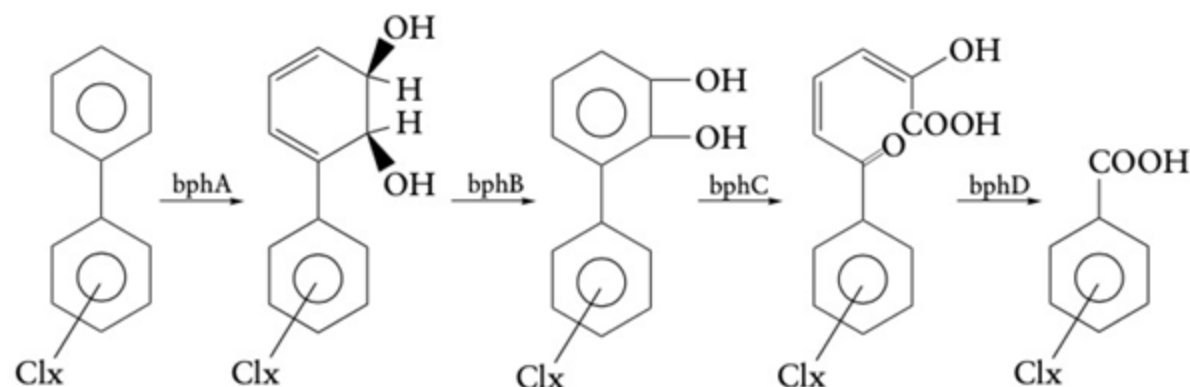
## The 12 Key POPs—the Dirty Dozen

| POP                | Use                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------|
| Aldrin             | crop insecticide (corn, cotton)                                                                   |
| Chlordane          | crop insecticide (vegetables, citrus, cotton, potatoes)<br>crop insecticide (cotton)              |
| DDT                |                                                                                                   |
| Dieldrin           | crop insecticide (cotton, corn)                                                                   |
| Endrin             | crop insecticide (cotton, grains)                                                                 |
| Heptachlor         | insecticide (termites and soil insects)                                                           |
| Hexa-chlorobenzene | fungicide for seed treatment                                                                      |
| Mirex              | insecticide (termites, fire ants)                                                                 |
| Toxaphene          | insecticide (livestock and crops)                                                                 |
| PCBs               | industrial chemical (heat exchange fluid for electrical transformers, paint and plastic additive) |
| Dioxins            | unintentionally produced during combustion                                                        |
| Furans             | unintentionally produced during combustion                                                        |

PCBs = Polychlorinated Biphenyls

### Polychlorinated biphenyls (PCBs)

members of the genus *Pseudomonas*, degrade PCBs in the presence of oxygen



The metabolic pathways used by these diverse microorganisms for PCB destruction are similar

### Aliphatic compounds

The reductive dehalogenation of aliphatic compounds has also been observed

For example, 1,1,1-trichloroethane (methylchloroform) can be reduced to the corresponding dichloro derivative



Anaerobic microorganisms reduce the chloro groups from the highly substituted biphenyls to give lightly chlorinated biphenyls, and the aerobic microorganisms destroy the lightly substituted biphenyls

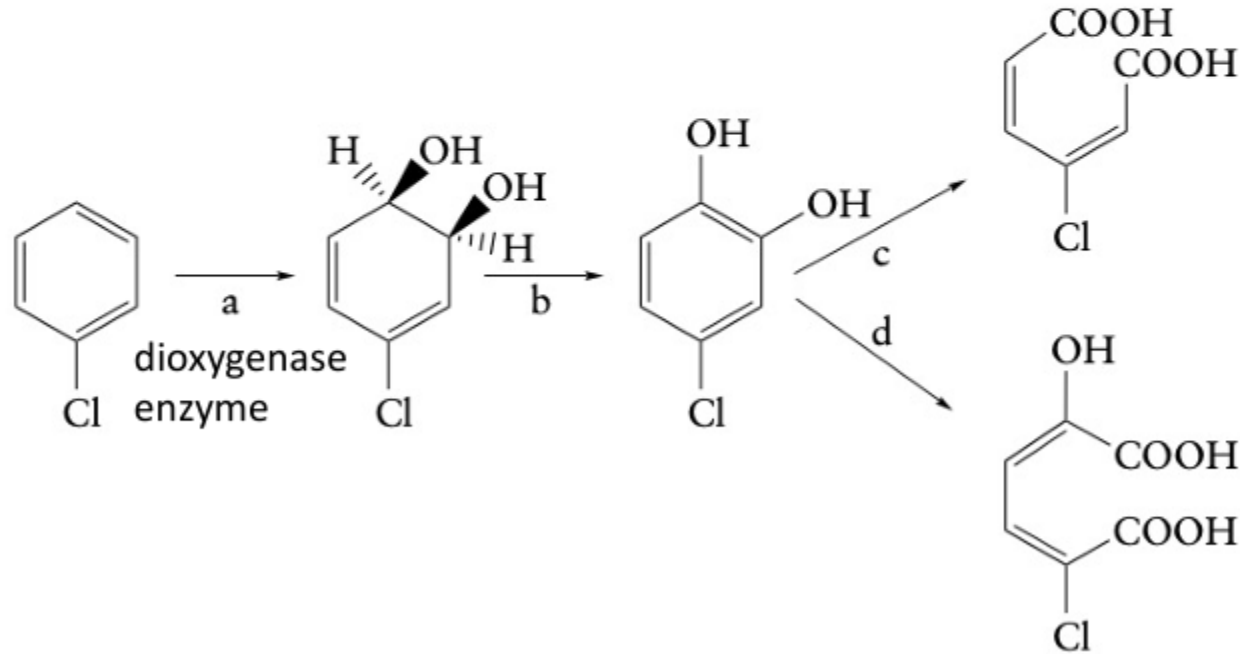


## Oxidative Degradation

Occur mainly with lightly chlorinated chloroorganics

because they require the attack of an electrophilic oxygen on an electron-rich center  
Such as an aromatic ring or a double bond

The delocalization of these electrons by electronegative chlorine atoms  
decreases the electron density on the aromatic ring;  
hence these compounds are less readily oxidized by molecular oxygen



readily degraded to simpler compounds and  
eventually to  $\text{CO}_2$ ,  $\text{H}_2\text{O}$ , and  $\text{Cl}^-$